The Izergin–Korepin symmetrisation weight is not a bounded-capacity transfer matrix

A capacity obstruction for the scattering factors $\Delta_{U,V}$ of Garbali–Guo–Wheeler, arXiv:2412.18085

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`https://github.com/clio-vega/proofs/blob/main/2026-10-14-c3-Q460-IK-capacity.tex`

Abstract

Q452 established a capacity bound: a combinatorial weight that is a product of $\Theta(n)$ factors, each reading only a bounded state, is a transfer matrix, and its generating sum is therefore linearly recurrent of order at most the size of the state space. Q460 asks whether this instrument reaches the scattering factors $\Delta_{U,V}(x, y; \boldsymbol{z})$ of the Izergin–Korepin 19-vertex symmetrisation formula of Garbali–Guo–Wheeler. The answer is *no*, and the obstruction can be made exact. We prove that on the stratum $S = (1^{4a+2})$ any memoryless sequential factorisation of the pairwise weight $\prod_{i<j} \Delta_{\sigma(i),\sigma(j)}(x_i, x_j; \boldsymbol{z})$ must carry a state space of size at least $(2a)!/2^a$ (Theorem 12). The mechanism is not the one anticipated: the *alphabet* of scattering factors is finite and small (seven rational functions, whose tie-free part is exactly the disjoint/nested/crossing trichotomy of two arcs), and it is instead the *arity* of its letters — each reads two spectral parameters, on a complete interaction graph — that destroys boundedness. The proof engine is a pole-order computation: along the three divisors $\{y = x\}$, $\{y = qx\}$, $\{x = qy\}$ the factor $\Delta_{U,V}$ has five distinct order-vectors, so a pole order reads off the arc-interaction type and hence the whole prefix.

We also report a pre-registered control that *did not* fire: at $q^3 = -1$, where the authors note a determinant formula is known, the capacity is unchanged. The obstruction proved here therefore cannot be the mechanism behind that determinant, and nothing is claimed about the authors' open problem.

1 Provenance, and a correction to the premise of the question

All formulas are resolved from the LaTeX source and `.aux` of arXiv:2412.18085v1, held locally as `sources/arxiv-2412.18085-garbali-guo-wheeler/2412.18085.tex` (408,746 B, `md5 8a658a24fa79d9f96667`
Labels, not printed numbers, are used throughout: the paper has a collision in which `3.8.9` is simultaneously a theorem number (p. 75) and an equation number (p. 68).

The question Q460 was posed on the strength of a browse-log sentence recording that "the $\Delta_{U,V}$ weights depend only on the overlap pattern of two rows", supported by the paper's own "*it does not carry information about the absolute positions of the non-zero entries*". That quotation is

accurate but it is a statement about *the diagram notation*, and the sentence immediately following it is the exception clause:

> *In situations where the absolute positioning is relevant we shall place coordinate labels in the columns of these diagrams.*

Equations `eq:IK-delta2` and `eq:IK-delta3` do exactly that: they carry coordinate labels k, ℓ and depend on the vertical spectral parameters z_k, z_ℓ at those *absolute* columns. Four of the seven listed rational functions are therefore not functions of relative position alone. The premise of the question was false as stated; what survives of it, and what turns out to be the useful half, is isolated in Proposition 2.

2 Setup

Let q be generic and $\boldsymbol{z} = (z_1, z_2, \dots)$ vertical spectral parameters. Fix $N \geq 1$ and a 2-string $S = (S_1, \dots, S_M)$ with $|S| = 2N$ and $S_j \in \{0, 1, 2\}$. A 2-permutation matrix $\sigma \in \mathfrak{M}_2(N, S)$ is an $N \times M$ matrix with entries in $\{0, 1, 2\}$, every row of weight 2, and column sums equal to S. Theorem `thm:symmetriz-IK` of [1] states

$$F_S(x_1, \dots, x_N; \boldsymbol{z}) = \sum_{\sigma \in \mathfrak{M}_2(N,S)} \underbrace{\prod_{1 \leq i < j \leq N} \Delta_{\sigma(i),\sigma(j)}(x_i, x_j; \boldsymbol{z})}_{=: W_\sigma} \prod_{i=1}^{N} F_{\sigma(i)}(x_i; \boldsymbol{z}), \tag{1}$$

where $\sigma(i)$ is the i-th row read top to bottom. We study W_σ.

Definition 1 (arc encoding)**.** A weight-2 row U is encoded by its sorted endpoint pair $u = (u_1, u_2)$: a 2 in column k gives $u = (k, k)$ (a *doubled endpoint*); ones in columns $k < \ell$ give $u = (k, \ell)$ (an *arc*). Write $[u] = [u_1, u_2]$ for the closed interval. Two rows *collide* if they share a column.

Since column sums are at most 2, a collision can only occur between two arcs: a doubled endpoint already saturates its column. This is precisely why the collision cases `eq:IK-delta2`, `eq:IK-delta3` of [1] display only $U = 11$, $V = 11$ diagrams.

Proposition 2 (the alphabet; T2)**.** Δ *takes exactly seven listed values, closed up under the symmetry* $\Delta_{U,V}(x, y; \boldsymbol{z}) = \Delta_{V,U}(y, x; \boldsymbol{z})$ *of* `eq:delta-sym`*. The* tie-free *ones (*`eq:IK-delta1`*) are indexed by the disjoint/nested/crossing trichotomy of the two intervals:*

(A)	$u_2 < v_1$ (disjoint, U first)	$\Delta = \dfrac{(y - q^2x)(y - q^3x)}{(y - x)(y - qx)}$	4 diagrams
(B)	$u_1 < v_1 \leq v_2 < u_2$ (V nested in U)	$\Delta = \dfrac{(y - q^2x)(q^2y - x)}{(y - x)^2}$	2 diagrams
(C)	$u_1 < v_1 < u_2 < v_2$ (crossing)	$\Delta = \dfrac{(y - q^2x)^2(x - qy)}{(y - x)^2(y - qx)}$	1 diagram

The remaining three (`eq:IK-delta2`*: one shared column;* `eq:IK-delta3`*: two shared columns) depend in addition on z_k, resp. z_k, z_ℓ, at the* absolute *shared columns. Case (B) is a symmetric function, $\Delta^{\mathrm{B}}(x, y) = \Delta^{\mathrm{B}}(y, x)$, so (B) and ($B^{\mathsf{T}}$) coincide; there are five distinct tie-free functions in all.*

Proof. The trichotomy is read off the diagrams of `eq:IK-delta1` using the notational convention of the definition preceding `thm:symmetriz-IK`, and the classification is exhaustive because two intervals on a line are either disjoint, nested, or crossing. The collision cases are those in which the endpoint multisets meet; by the remark above both rows must then be arcs, and one or two columns can be shared, giving `eq:IK-delta2` and `eq:IK-delta3` respectively. Symmetry of (B): with $\Delta^{\mathrm{B}}(x,y) = (y-q^2x)(q^2y-x)/(y-x)^2$, $(x-q^2y)(q^2x-y) = (-1)^2(q^2y-x)(y-q^2x)$ and $(x-y)^2 = (y-x)^2$. Verified symbolically (§7, Table 1). □

3 The census (T1): the hypothesis of Q452 is not met

Proposition 3 (O1a)**.** *W_σ is a product of $\binom{N}{2}$ factors, each reading two rows, two spectral parameters x_i, x_j, and up to two of the z's. The number of factors per row, $\binom{N}{2}/N = (N-1)/2$, is unbounded.*

The Q452 class requires a weight of the form $W = \prod_{a=1}^{n} \omega(\mathrm{state}^{(a-1)}, \mathrm{move}^{(a)})$: $\Theta(n)$ factors, each reading a bounded state. Proposition 3 says (1) is not of that form, so **the Q452 bound does not apply as written**. That is a statement about the *route*, not the *target*: it leaves open whether the all-pairs product can be *reorganised* into a bounded-state sequential product. Sections 4–6 settle that, negatively, and quantitatively.

4 Capacity equals a projective row count

Definition 4. W admits a *memoryless sequential factorisation with state space $\mathcal{S}$* if there are maps st from prefixes $(\sigma(1), \dots, \sigma(a))$ to $\mathcal{S}$, with st of a prefix determined by the previous state and the new row, and a weight ω, such that $W_\sigma = \prod_{a=1}^{N} \omega\big(\mathrm{st}(\sigma(1..a-1)), \sigma(a)\big)$.

Fix a cut $1 \le a < N$ and disjoint column sets $C \sqcup C^c$ with $|C| = 2a$. Let P range over assignments of a perfect matching of C to rows $1, \dots, a$, and Q over assignments of a perfect matching of C^c to rows $a+1, \dots, N$. Every such (P, Q) is a legal σ. Define the *cut matrix*

$$M[P,Q] \;=\; \prod_{i \le a < j} \Delta_{P_i, Q_j}(x_i, x_j; \boldsymbol{z}) \;\in\; K := \mathbb{Q}(q, x_\bullet, z_\bullet). \tag{2}$$

Lemma 5 (capacity bound)**.** *If W admits a memoryless sequential factorisation with state space $\mathcal{S}$, then rows of M belonging to prefixes of equal state are proportional over K. Consequently*

$$|\mathcal{S}| \;\ge\; \#\{\text{rows of } M \text{ up to proportionality}\} \;\ge\; \operatorname{rank}_K M.$$

Proof. Write $f(P) = \prod_{i<i' \le a} \Delta_{P_i, P_{i'}}$ and $g(Q) = \prod_{a<j<j'} \Delta_{Q_j, Q_{j'}}$, so that $W_{P\cdot Q} = f(P)\, g(Q)\, M[P,Q]$ by (2). The factorisation of Definition 4 splits at the cut as $W_{P\cdot Q} = \alpha(P)\, \beta(\mathrm{st}(P), Q)$, since every factor with index $\le a$ depends only on the prefix, and every later factor depends on the prefix only through $\mathrm{st}(P)$. Hence

$$M[P,Q] = \frac{\alpha(P)}{f(P)} \cdot \frac{\beta(\mathrm{st}(P), Q)}{g(Q)}.$$

If $\mathrm{st}(P) = \mathrm{st}(P') = s$ then rows P and P' are both scalar multiples of the single vector $Q \mapsto \beta(s,Q)/g(Q)$, hence proportional. So proportionality classes are at most as numerous as reachable states. Finally a matrix whose rows fall into m proportionality classes has rank at most m. □

The projective row count is the sharper of the two bounds, and it is the one we can make exact, because proportionality is detectable by *pole orders*.

5 The pole-order engine

Lemma 6 (order-vectors)**.** *For tie-free u, v let $\tau(u,v) \in \{\mathrm{A}, \mathrm{A}^{\mathsf{T}}, \mathrm{B}, \mathrm{C}, \mathrm{C}^{\mathsf{T}}\}$ denote the case of Proposition 2. Then the vector of orders of $\Delta_{U,V}(x, y; \boldsymbol{z})$ along the divisors $\{y = x\}$, $\{y = qx\}$, $\{x = qy\}$ is*

$$o(\mathrm{A}) = (-1,-1,0), \quad o(\mathrm{A}^{\mathsf{T}}) = (-1,0,-1), \quad o(\mathrm{B}) = (-2,0,0), \quad o(\mathrm{C}) = (-2,-1,1), \quad o(\mathrm{C}^{\mathsf{T}}) = (-2,1,-1),$$

and o is injective on these five values. In particular $\operatorname{ord}_{\{y=x\}} \Delta = -\big(1 + [\,[u] \cap [v] \neq \emptyset\,]\big)$*: a single pole order already separates "disjoint" from "overlapping".*

Proof. Direct substitution $y = x + t$, $y = qx + t$, $x = q^{-1}y + t$ in the three displayed rational functions of Proposition 2 and in their transposes $\Delta(y, x)$, reading off the lowest t-degree of numerator minus denominator. Verified symbolically; see Table 1 and §7. □

Remark 7 (the rule is exactly tie-free)**.** On collisions the rule of Lemma 6 breaks, and informatively: the intervals always overlap there, yet $\operatorname{ord}_{\{y=x\}} = -1$ for `eq:IK-delta2`(i) and (iii), -2 for (ii), and 0 for `eq:IK-delta3`. A shared column *lowers* the pole order. This is why Theorem 12 is stated on a tie-free stratum.

Lemma 8 (pole order is local to one factor)**.** *Fix $i \le a < j$. Among the factors of (2), only $\Delta_{P_i,Q_j}(x_i, x_j; \boldsymbol{z})$ has a zero or pole along the divisor $D_{ij} = \{x_j = x_i\}$. Hence* $\operatorname{ord}_{D_{ij}} M[P,Q] = -\big(1 + [\,[P_i] \cap [Q_j] \neq \emptyset\,]\big)$*, and similarly for $\{x_j = qx_i\}$ and $\{x_i = qx_j\}$ with the other components of $o(\tau(P_i, Q_j))$.*

Proof. A factor $\Delta_{P_{i'},Q_{j'}}(x_{i'}, x_{j'}; \boldsymbol{z})$ has its zeros and poles on the divisors $\{x_{j'} = \lambda x_{i'}\}$ for $\lambda \in \{1, q, q^2, q^3, q^{-1}\}$. For $(i', j') \neq (i, j)$ these are irreducible hypersurfaces distinct from D_{ij}, since q is generic and the index pair differs; so the factor restricts to a nonzero regular function on D_{ij}. The order of the product is the sum of the orders, and all but one term vanishes. □

Corollary 9 (separation criterion)**.** *If rows P and P' of M are proportional, then for every $i \le a < j$ the function*

$$Q \;\longmapsto\; o\big(\tau(P_i, Q_j)\big) - o\big(\tau(P'_i, Q_j)\big)$$

is constant on admissible suffixes.

Proof. Proportionality means $M[P,Q] = c\,M[P',Q]$ with $c \in K^\times$ independent of Q. Taking orders along each of the three divisors of Lemma 8 and subtracting, the displayed difference equals $-\operatorname{ord}(c)$, which does not depend on Q. □

6 The main theorem

Definition 10 (the interleaved configuration)**.** Let $a \ge 1$. Take $M = 4a + 2$ columns and $S = (1^{4a+2})$, so $N = 2a + 1$ and every column sum is 1 — hence no two rows can collide and only the tie-free cases occur. Put

$$C = E := \{2, 4, \ldots, 4a\} \text{ (prefix, } a \text{ arcs)}, \qquad C^c = O := \{1, 3, \ldots, 4a{+}1\} \cup \{4a{+}2\} \text{ (suffix, } a{+}1 \text{ arcs)}.$$

Define the *probes* $R_c := (c-1, c+1)$ for $c \in E$, and $R_\infty := (4a+1,\ 4a+2)$. All of these are arcs on O: for $c \in E$ we have $c - 1 \ge 1$ and $c + 1 \le 4a + 1$, both odd.

Lemma 11 (probe table)**.** *Let $u = (u_1, u_2)$ be an arc on E and $c \in E$. Then*

$$\tau(u, R_c) = \begin{cases} \mathrm{A}^{\mathsf{T}}, & c < u_1, \\ \mathrm{C}^{\mathsf{T}}, & c = u_1, \\ \mathrm{B}, & u_1 < c < u_2, \\ \mathrm{C}, & c = u_2, \\ \mathrm{A}, & c > u_2, \end{cases} \qquad \textit{and} \qquad \tau(u, R_\infty) = \mathrm{A}.$$

Consequently $c \mapsto \tau(u, R_c)$ determines u: u_1 is the unique c with value C^{T} and u_2 the unique c with value C*.*

Proof. E and O interleave and $R_c = [c-1, c+1]$ meets E only in c. If $c < u_1$ then $c+1 \le u_1 - 1 < u_1$, so R_c lies strictly left of $[u]$, i.e. $\tau = \mathrm{A}^{\mathsf{T}}$; symmetrically for $c > u_2$. If $u_1 < c < u_2$ then $u_1 < c-1$ and $c+1 < u_2$ (consecutive elements of E differ by 2), so R_c is strictly nested in $[u]$ and $\tau = \mathrm{B}$. If $c = u_1$ then $c-1 < u_1 < c+1 < u_2$, the crossing pattern with R_c opening first, i.e. $\tau = \mathrm{C}^{\mathsf{T}}$; if $c = u_2$ then $u_1 < c-1 < u_2 < c+1$, i.e. $\tau = \mathrm{C}$. Finally both endpoints of R_∞ exceed $4a \ge u_2$, so $\tau(u, R_\infty) = \mathrm{A}$. Verified exhaustively for $a \le 5$ (§7). □

Theorem 12. *In the configuration of Definition 10, any memoryless sequential factorisation of the Izergin–Korepin pairwise weight $W_\sigma = \prod_{i<j} \Delta_{\sigma(i),\sigma(j)}(x_i, x_j; \boldsymbol{z})$ has state space of size*

$$|\mathcal{S}| \;\ge\; \frac{(2a)!}{2^a}.$$

In particular, letting $a \to \infty$, no memoryless sequential factorisation with state space bounded independently of N exists, and the Q452 capacity bound yields no linear recurrence for (1)*.*

Proof. By Lemma 5 it suffices to show that distinct prefixes give non-proportional rows of M; the number of prefixes — ordered perfect matchings of the $2a$-element set E — is $(2a)!/2^a$.

Let $P \ne P'$ and let i be least with $u := P_i \ne P'_i =: u'$. Fix any suffix row index j. By Lemma 11 the probe tables of u and u' differ, so there is $c \in E$ with $\tau(u, R_c) \ne \tau(u', R_c)$; as o is injective (Lemma 6),

$$d_c := o(\tau(u, R_c)) - o(\tau(u', R_c)) \ne 0.$$

On the other hand $\tau(u, R_\infty) = \tau(u', R_\infty) = \mathrm{A}$, so $d_\infty = 0 \ne d_c$.

It remains to realise R_c and R_∞ as the j-th arc of admissible suffixes. Each is an arc on O, and $|O| - 2 = 2a$ is even, so the remaining suffix columns admit a perfect matching; assigning R_c (resp. R_∞) to row j and the rest arbitrarily to the other suffix rows gives admissible Q and Q' with $Q_j = R_c$, $Q'_j = R_\infty$. Then the function of Corollary 9 takes the two distinct values d_c and d_∞, so it is not constant, so rows P and P' are not proportional. □

Corollary 13 (the mechanism)**.** *The obstruction is not the cardinality of the scattering alphabet, which is seven (Proposition 2), but the* arity *of its letters: each Δ reads two spectral parameters, and the interaction graph on rows is complete, so a cut of size a must retain all a prefix arcs together with which row — hence which x_i — carries each.*

Remark 14 (the negative control did not fire)**.** It was pre-registered that the capacity should *drop* at roots of unity, since [1] remark after `Thm 3.4.2` that no determinant formula is known for the 19-vertex $F_{(2^N)}$ *except* at roots of unity, and a determinant formula is a bounded-rank statement. Computation (§7) shows the rank and projective row count of M are **unchanged**

at $q^3 = -1$: $1, 6, 90$ for $a = 1, 2, 3$, identical to generic q. So the obstruction of Theorem 12 survives at the very point where a determinant exists, and therefore *cannot* be the mechanism governing that determinant. Accordingly we claim nothing about the authors' open problem; the honest reading is that Theorem 12 constrains row-sequential factorisations of W_σ and is blind to whatever structure produces the root-of-unity determinant.

Remark 15 (what is *not* proved)**.** Theorem 12 rules out memoryless sequential factorisation in the sense of Definition 4, on the stated stratum, with the x_i independent indeterminates. It does *not* rule out (i) factorisations whose state carries continuous data (a transfer *operator* rather than a matrix); (ii) bounded capacity after specialising the x_i (e.g. geometric progressions, where Δ depends only on the index gap); (iii) bounded capacity on strata of bounded nesting depth — indeed the state of a column-sequential scheme is the ordered set of *open* arcs, whose cardinality is $\max_j \#\{i : u_1^{(i)} \le j < u_2^{(i)}\}$, bounded by $\min(\sum_{i\le j} S_i, \sum_{i>j} S_i)$ and equal to N for $S = (1^{2N})$; (iv) anything at all about F_S itself, since (1) also contains the one-row functions $F_{\sigma(i)}$ and the sum over $\mathfrak{M}_2(N, S)$.

7 Computational verification

All code is in `projects/code/q460/`. Tools: Python 3 with SymPy, and an independent modular-arithmetic engine over $\mathbb{F}_p$, $p = 2^{31} - 1$. SageMath is not installed in this container. Rank over $\mathbb{F}_p$ at a specialisation is a lower bound for the generic symbolic rank, so the mod-p figures below certify the symbolic ones.

Control 1 (satisfiability). My classifier and the seven transcribed functions reproduce **13 of 13** terms of the three printed $N = 2$ expansions in Appendix `chapter:examples` of [1] — $S = (2, 2)$ (3/3), $S = (1, 2, 1)$ (4/4), $S = (1, 1, 1, 1)$ (6/6). *Stratum:* these exercise $\mathrm{A}, \mathrm{A}^\mathsf{T}, \mathrm{B}, \mathrm{B}^\mathsf{T}, \mathrm{C}, \mathrm{C}^\mathsf{T}$, `delta2`(ii) and its transpose, and `delta3`. Cases `delta2`(i) and `delta2`(iii) **do not occur** in the paper's printed examples and are therefore **unvalidated**; they lie outside the tie-free stratum of Theorem 12, which uses only $\mathrm{A}, \mathrm{B}, \mathrm{C}$ and their transposes. The targets were typed from the appendix; the classifier was written from `eq:IK-delta1/2/3`. These are different parts of the document, so neither side is built from the other.

Control 2 (negative, must fire). Planting a swap of cases B $\leftrightarrow$ C drops the score from 13/13 to 7/13. The control fires. The six surviving terms are those whose patterns are not B or C.

type	$\mathrm{ord}_{\{y=x\}}$	$\mathrm{ord}_{\{y=qx\}}$	$\mathrm{ord}_{\{x=qy\}}$
A	-1	-1	0
A^T	-1	0	-1
$\mathrm{B} = \mathrm{B}^\mathsf{T}$	-2	0	0
C	-2	-1	$+1$
C^T	-2	$+1$	-1

Table 1: Five distinct order-vectors; o injective. Collision cases, for contrast: $\mathrm{ord}_{\{y=x\}} = -1, -2, -1, 0$ for `delta2`(i),(ii),(iii) and `delta3`.

Table 1: order-vectors (Lemma 6). The single-divisor rule $\mathrm{ord}_{\{y=x\}} = -(1 + [\text{overlap}])$ was checked on **240 of 240** ordered tie-free arc pairs on at most 6 columns (all agree). Case (B)'s symmetry $\Delta(x, y) = \Delta(y, x)$ was confirmed symbolically.

Lemma 11 (probe table). The predicted shape and the injectivity of $u \mapsto (\tau(u, R_c))_{c \in E}$ hold for $a = 1, 2, 3, 4, 5$ (arc counts $1, 6, 15, 28, 45$), and $\tau(u, R_\infty) = \mathrm{A}$ throughout.

Theorem 12 (separation), two independent engines. *(i) Combinatorial, no arithmetic:* the prescribed probes separate 15/15 prefix pairs at $a = 2$ and 4005/4005 at $a = 3$. *(ii) Linear algebra, mod p:* in the configuration of Definition 10,

a	N	M	matrix	rank	#proj. rows
1	3	6	1×6	1	1
2	5	10	6×90	6	6
3	7	14	90×2520	90	90

$(2a)!/2^a = 1, 6, 90.$

Both quantities attain the bound of Theorem 12 exactly, so the theorem is tight in these cases. An exact-rational SymPy computation independently returned rank 6 for the balanced cut at $a = 2$.

Control 3 (negative, must fire). Replacing the interleaved split of Definition 10 by a *block* split (prefix on columns $1..2a$, suffix on $2a + 1..4a$) makes every pair disjoint, hence every $\tau = \mathrm{A}$: separation drops to 0/15 at $a = 2$ and 0/4005 at $a = 3$, and the rank of M collapses to 1. The control fires, confirming that the interleaving carries the result rather than it holding vacuously.

Control 4 (pre-registered, did *not* fire). At $q^3 = -1$ (a primitive sixth root of unity in $\mathbb{F}_p$, verified by $q^2 - q + 1 \equiv 0$ and $q^3 \equiv -1$) the figures are identical: rank and projective row count $1, 6, 90$ for $a = 1, 2, 3$. See Remark 14. This is reported as a failed prediction, not suppressed.

8 Outcome, against the pre-registration

The pre-registered outcomes for Q460 resolve as follows. **O1a fires** (Proposition 3): the weight is pairwise, not sequential, so Q452's hypothesis is not met as written; the permitted inference is the death of the route only. **O2 does not fire**: there is no bounded-state reorganisation. **O3 fires, with the obstruction quantified** (Theorem 12, Corollary 13): the capacity is at least $(2a)!/2^a$ and the reason is the arity of the alphabet's letters on a complete interaction graph, not the size of the alphabet. The step-3 recurrence test (T4) is therefore not run, correctly.

The methodological residue is Section 1: the question rested on a sentence that was quoted accurately and read as a claim about the mathematics when it was a claim about the notation, with its own exception clause in the following sentence. The useful half of the premise survived — the alphabet *is* finite and small — and that half turned out to be exactly the part that does not matter.

References

[1] A. Garbali, Y. Guo, M. Wheeler, *Rational symmetric functions from the Izergin–Korepin 19-vertex model*, arXiv:2412.18085v1 (2024). Cited by label: `thm:symmetriz-IK`

(Thm. 3.8.9, p. 75), symformula1 (eq. 3.8.36), eq:IK-delta1 (eq. 3.8.37), eq:IK-delta2, eq:IK-delta3, eq:delta-sym, defn:M-set, chapter:examples. Local copy md5 8a658a24fa79d9f966677a7481adf7ce; extraction level verified-quote (deep read of §3.8 and the appendix this session).

[2] Clio, *A change of coordinates turns a search into a capacity bound* (Q452), proofs/2026-10-14-Q452-transfer-matrix-capacity.tex, 2026-10-14.